

Simulating 3D Thermal Fluid Dynamics in Data Centers with Soft-Constrained Physics-Informed Graph Neural Network

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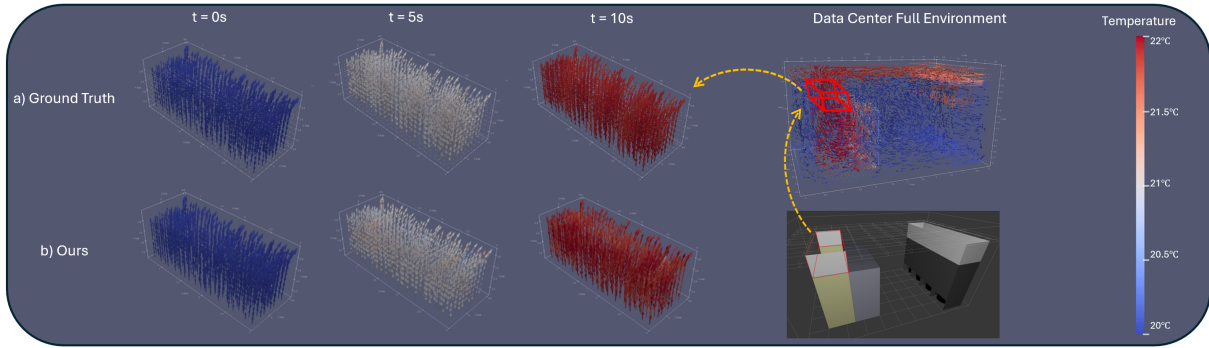


Figure 1: Simulation of temperature and airflow in the hot aisle region of a 3D data center. The full environment is shown on the right, with the hot aisle region (highlighted in red boxes) extracted for analysis; the orange arrow indicates its selection. The simulated result is visualized on the left, comparing (a) the ground truth CFD simulation and (b) our SC-PIGNN3D prediction at three time-steps ($t = 0$ s, $t = 5$ s, and $t = 10$ s). Arrows denote flow velocity, with length proportional to magnitude and color indicating temperature. This visualization captures both the temporal evolution of airflow and the consistency between our model predictions and the ground truth.

Abstract

Data-driven methods have emerged as efficient alternatives to traditional computational fluid dynamics (CFD) methods for fluid simulation, offering faster inference and lower computational cost. However, most are limited to 2D data or neglect key quantities such as temperature, restricting their use in 3D thermal analysis. We introduce a **soft-constrained physics-informed graph neural network**, or **SC-PIGNN3D**, for transient 3D simulation of velocity, pressure, and temperature fields in data center environments. To address slow convergence in large-scale physics-informed GNNs, we adopt a dual-processor architecture that separates learning into task-specific modules. Experiments show that our SC-PIGNN3D

achieves accurate, efficient simulations while dual processor approach is able to cut training time by nearly half.

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1 Introduction

Efficient thermal management in data centers is vital due to high cooling energy use. While CFD offers accurate airflow and temperature simulations, it is too slow for complex and transient cases. Emerging data-driven models like GNNs and PINNs [Hernández et al. 2022] capture flow dynamics with physical laws, yet most either exclude temperature prediction or remain constrained to 2D simulations [Shao et al. 2023].

In this paper, we present a **soft-constrained physics-informed graph neural network**, or **SC-PIGNN3D** in short, a novel data-driven framework designed for 3D dynamic airflow simulation in

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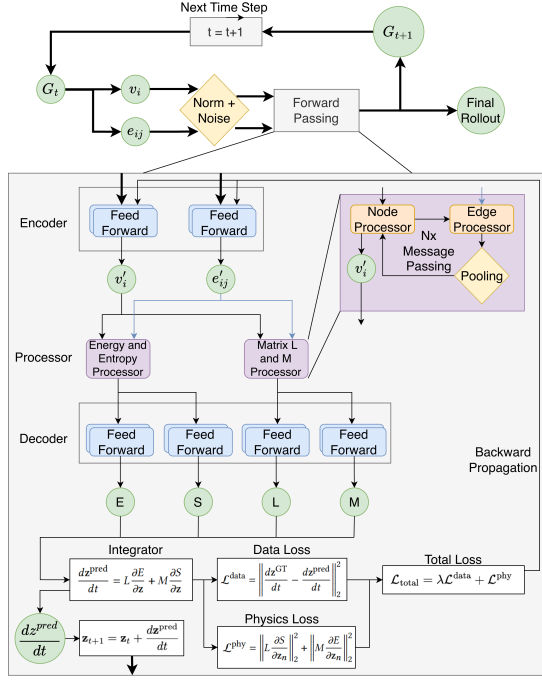


Figure 2: Network structure of the soft-constrained physics-informed graph neural network.

3D data centers. Our model predicts high-dimensional flow fields, including velocity, pressure, and temperature, and further improves convergence speed through the integration of specialized processors for output predictions. Since one of the most critical challenges in thermal management is timely **hotspot detection**, we tested our model in the **hot aisle region** of a data center shown on the left of Figure 1, where heated airflow from server racks is recirculated to air cooling units.

2 Our Approach

Our SC-PIGNN3D framework is shown in Figure 2. To construct the graph, we adopt a mesh-based representation. Unlike the particle-based approach, the simulation domain is discretized into meshes, and each mesh point corresponds to a node carrying the local physical state, which we will later refer to as state variables. Edges are defined on the basis of spatial adjacency, capturing geometric relations between neighboring nodes. The simulation state at time step t is represented by a graph $G_t = (v_i, e_{ij})$, where node v_i contains state variables (velocity, pressure, temperature) and edge e_{ij} encodes geometric relations (relative positions). The state G_t is normalized and perturbed with small noise before processing. An **Encoder** maps raw features to latent embeddings via MLPs, $e_{ij} \rightarrow e'_{ij}$, $v_i \rightarrow v'_i$. A dual-processor GNN **Processor** with N message-passing blocks updates edges and nodes via $(e'_{ij}, v'_i, v'_j) \rightarrow e'_{ij}$ and $(v'_i, \Phi(e'_{ij})) \rightarrow v'_i$, where Φ is permutation-invariant pooling. One processor predicts scalars (Energy E , Entropy S), the other predicts tensor operators (L , M). The **Decoder**

outputs $E = \text{MLP}_E(v'_i)$, $S = \text{MLP}_S(v'_i)$, $L = \text{MLP}_L(v'_i)$, $M = \text{MLP}_M(v'_i)$.

The network computes the temporal derivative of the state, denoted as $\frac{dz^{\text{pred}}}{dt}$, using the GENERIC formulation operators [Hernández et al. 2022]: $\frac{dz^{\text{pred}}}{dt} = L \frac{\partial E}{\partial z} + M \frac{\partial S}{\partial z}$. This predicted derivative describes how the state variables evolve over time. The next state is then obtained by explicit Euler integration, $z_{t+1} = z_t + \frac{dz^{\text{pred}}}{dt}$, yielding the updated graph state $G_{t+1} = (z_{t+1}, t + 1)$. This recurrent rollout continues until the full simulation horizon is reached.

Loss is calculated on every epoch, and the total loss is defined as $\mathcal{L}_{\text{total}} = \lambda \left\| \frac{dz^{\text{GT}}}{dt} - \frac{dz^{\text{pred}}}{dt} \right\|_2^2 + \left\| L \frac{\partial S}{\partial z_n} \right\|_2^2 + \left\| M \frac{\partial E}{\partial z_n} \right\|_2^2$. The first term, scaled by the weight λ , is the data loss part as shown on the right of Figure 1. It is a relative loss calculated by comparing the difference between the ground truth and the predicted value. The latter terms act as the physics loss, which is determined by GENERIC formulation soft constraints, ensuring that the learned operators respect thermodynamic laws. Together, these components form the total loss, which is back-propagated.

3 Experimental Results

Table 1: Experiment results for different setups of the soft-constrained physics-informed graph neural network

Configuration	Training Time (Hours)	VRAM Cost (GB)	Relative L2 Error (%)
SC-PIGNN3D (2k Epoch)	47(± 2)	35	V = 16.3 $\pm$ 1.25 P = 7.68 $\pm$ 0.6 T = 0.337 $\pm$ 0.078
SC-PIGNN3D + Dual Processor (1k Epoch)	27(± 1)	43	V = 15.7 $\pm$ 2.93 P = 8.29 $\pm$ 1.42 T = 0.317 $\pm$ 0.145

The simulated 3D data center environment, modeled using the *Cadence Reality Digital Twin Platform*, includes air-conditioning units, GPU server racks and hot aisle–cold aisle containment zones. We focus on the **hot aisle tunnel** region, represented by 4,000 unstructured 3D points with coordinates (x, y, z) , velocity components, pressure and temperature. The points are classified as **driving inlets, outlets, walls, or interior flow points**, with boundary points enforcing training constraints. The simulation covers 10 seconds at 0.05-second intervals, yielding 200 time steps under constant server workload.

Training is performed on a single NVIDIA A100 GPU with two hidden layer MLPs (128 units), learning rate 10^{-4} , noise variance 4×10^{-4} , and loss-balancing weight $\lambda = 10$. Table 1 summarizes performance, showing that the dual-processor design achieves comparable accuracy while reducing training time by 42%.

4 Conclusion future work

The SC-PIGNN3D with a dual-processor architecture delivers faster convergence and accurate simulation in airflow and thermal field simulations. Future work will expand datasets for robustness, benchmark against classical GNNs and state-of-the-art surrogate models,

and extend evaluations to diverse airflow configurations and operating conditions.

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